Credit risk: ranking to policy

To: DVP - Analytics | Date: 26 October 2026

Decision requested

Sponsor a controlled validation study on actual Indian MSME data. Retain the interpretable scorecard as a governance reference and assess the boosting challenger before any lending deployment.

Problem and evidence

The project ranks repayment difficulty on 307,511 labeled Home Credit applications. SQL joins aggregate bureau, prior applications, installments, POS and card history into 168 candidate features. The observed TARGET rate is 8.07%. This is consumer lending proxy data, not NeoGrowth or verified Indian MSME data.

The scorecard uses 20 training-selected WoE features. A LightGBM challenger is tuned using training-only cross-validation. A separate policy partition sets the cut-off; neither test nor holdout labels set preprocessing, model parameters or the recommended policy. The holdout contains 61,503 applications.

Untouched random holdout	Scorecard	LightGBM
AUC	0.7519	0.7813
Gini	0.5039	0.5626
KS	37.64%	42.81%
Brier score (lower is better)	0.06856	0.06653
Observed / predicted TARGET rate	8.07% / 8.07%	8.07% / 8.06%

What the findings support

Both models show measurable ranking ability on the random holdout. LightGBM's holdout AUC difference from the scorecard is 0.0294; the paired bootstrap interval is [0.0265, 0.0326]. Scorecard score PSI is 0.00016 versus training. A random split cannot establish time stability.

Source: reports/metrics.json, SQL artifacts and fixed-seed executed code. TARGET denotes repayment difficulty; exact overdue/timing thresholds are withheld by the dataset provider. SHAP explanations show associations, not causes.

Policy economics and controls

Illustrative cut-off policy

Approve at score >= 545. This cut-off maximizes assumed net benefit on the policy partition subject to its minimum approval constraint. On untouched holdout, approval is 79.24%; accepted TARGET rate is 4.96%, versus 8.07% baseline. That is an estimated retrospective relative reduction of 38.59%, while rejecting 18.08% of observed good applicants.

Editable assumption / estimate	Value
Annual application count	100,000
Average loan / LGD / good-loan net margin	Rs 500,000 / 60.00% / 8.00%
Annual gross loss avoided (ILLUSTRATIVE)	Rs 124.34 crore
Annual good-loan margin foregone (ILLUSTRATIVE)	Rs 66.47 crore
Annual net benefit (ILLUSTRATIVE)	Rs 57.86 crore

Gross avoided loss = rejected bad share x annual applications x loan size x LGD. Net benefit subtracts rejected good share x annual applications x loan size x margin. Estimates assume unchanged applicant mix and identical exposure; they are not realized savings or a currency conversion.

Risks and implementation gates

There is no verified application chronology: this is random holdout validation. Selection bias, reject inference and label maturity remain unresolved. Calibration is measured, not corrected. Age, geography and occupation may encode sensitive proxies despite excluding gender. The age WoE coefficient reverses direction after adjustment and needs governance review.

Before piloting: reconstruct point-in-time MSME features and a documented target horizon; run temporal and segment validation; assess calibration and fairness; validate economic assumptions and approval capacity; establish independent review, score/feature drift monitoring, override logging and rollback controls.

SIMULATED early-warning extension

A separate artificial merchant-sales process produces SIMULATED precision 45.55%, recall 100.00% and median lead time 16 SIMULATED days among detected artificial events. The process deliberately builds in declining sales. These figures demonstrate alert logic and cannot establish real merchant performance.

Interactive dashboard and CSV export accompany this memo. Full assumptions and limitations: reports/LIMITATIONS.md. All quantitative claims are generated from reports/metrics.json.